

December 2024

B. Tech. 1st SEMESTER,

Mathematics-I (Calculus and Linear Algebra) (BSC-103E / BSCH-103E)

Time: 3 Hours

Max. Marks:75

- Instructions:**
1. It is compulsory to answer all the questions (1.5 marks each) of Part -A in short.
 2. Answer any four questions from Part -B in detail.
 3. Different sub-parts of a question are to be attempted adjacent to each other.

PART -A

Q1 (a) Evaluate

(1.5)

$$\int_0^1 x^{11}(1-x)^5 dx.$$

(b) Prove that $\Gamma(n+1) = n \Gamma(n)$, where $n > 0$. (1.5)

(c) State Rolle's theorem. Give its geometric interpretation. (1.5)

(d) Evaluate (1.5)

$$\lim_{x \rightarrow 0} \frac{e^{ax} - e^{bx}}{x}.$$

(e) Find the rank of the matrix (1.5)

$$\begin{bmatrix} 0 & 1 & 2 & 1 \\ 1 & 2 & 3 & 1 \\ 3 & 1 & 1 & 3 \end{bmatrix}.$$

(f) Find (1.5)

$$\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix}.$$

(g) Let $V = \mathbb{R}^2 = \{(a, b) : a, b \in \mathbb{R}\}$. Show that $W = \{(a, 0) : a \in \mathbb{R}\}$ is a subspace of V . (1.5)(h) Test whether the map $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(x, y, z) = (x, 0, 0)$ for all $(x, y, z) \in \mathbb{R}^3$ is linear. (1.5)(i) Find the eigenvalues of the matrix $\begin{bmatrix} 4 & 1 \\ 3 & 2 \end{bmatrix}$. (1.5)(j) If $A = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$, then find the eigenvalues of $A^2 - 2A + I$. (1.5)**PART -B**Q2 (a) Find the equation of the evolute of the parabola $y^2 = 4ax$. (7)(b) Find the volume generated when the area bounded by the parabolas $y^2 = 4 - x$ and $y^2 = 4 - 4x$ is revolved about the y-axis. (8)

Q3 (a) Evaluate

$$\lim_{x \rightarrow 0} \left(\frac{a^x + b^x + c^x + d^x}{4} \right)^{1/x}$$

where a, b, c and d are positive numbers.

(7)

(b) Show that

$$\log_e(1 + e^x) = \log_e 2 + \frac{x}{2} + \frac{x^2}{8} - \frac{x^4}{192} + \dots$$

and hence deduce that

$$\frac{e^x}{1 + e^x} = \frac{1}{2} + \frac{x}{4} - \frac{x^3}{48} + \dots$$

(8)

Q4 (a) Solve the following system of equations

$$x + y + z = 4$$

$$x - y + z = 0$$

$$2x + y + z = 5$$

(7)

(b) For what values of k , the equations

$$x + y + z = 1$$

$$2x + y + 4z = k$$

$$4x + y + 10z = k^2$$

(8)

have

(i) a unique solution,

(ii) infinitely many solutions,

(iii) no solution.

Q5 (a) Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be a linear transformation defined on the basis

$T(1, 0, 0) = (0, 0)$, $T(0, 1, 0) = (1, 1)$ and $T(0, 0, 1) = (1, -1)$.

Compute $T(4, -1, 1)$. Determine the nullity and rank of T .

(7)

(b) Find the matrix of the linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by

$$T(x, y) = (2x - y, 3x + 4y, x)$$

with respect to the standard bases of $\mathbb{R}^2$ and $\mathbb{R}^3$.

(8)

Q6 (a)

Find the eigenvalues and eigenvectors of the matrix $\begin{bmatrix} 3 & -4 & 4 \\ 1 & -2 & 4 \\ 1 & -1 & 3 \end{bmatrix}$.

(7)

(b) Find an orthonormal basis of the inner product space $\mathbb{R}^3$ over $\mathbb{R}$ with standard inner product, given the basis $B = \{(1, 0, 1), (0, 1, 1), (1, 3, 3)\}$ using Gram-Schmidt orthogonalization process. Also, find the Fourier coefficients of the vector $(1, 1, 2)$ relative to orthonormal basis.

(8)

Q7 (a) Prove that $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$.

(7)

(b) Diagonalize the matrix $A = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 3 & -1 \\ 1 & -1 & 3 \end{bmatrix}$ by means of an orthogonal transformation.

(8)